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Handoff —¿—°—”—µÑ: Recognition of cash receipts

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Project Overview

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— ÑÑ—,Ñ—µ—”ÑÑÑ—° —, —¿—°—„—¿—»—°—„—‰ (highâ level)

— Ñ—‰—¾—†—‰Ñ—µ—”—¾…—¿—¾—‰—µ—‰ÑÑ
- bot.py: Telegram UI/—»—¾—‡—,—”—°, —¿Ñ—,Ñ—… Ñ—¾Ñ—¾Ñ—µ—¶—,—…Ñ â —¾·—‰—¾Ñ—¾Ñ—¾â / â —·—»—,—‰
- ocr.py: —¿—°—„—¿—»—°—„—‰ Ñ—ºÑ—¿—¾—•—‰—°—†—°—‰—,Ñ (—¿Ñ—µ·—¾—Ñ—°———¾Ñ—”—° â QR â LLM v
Ñ—¾…—‰—µ—‰—,ÑÑ, —…—µÑ—°·—°—‰—‰Ñ—µ_meta.
- receipt_qr.py: ÑÑ—µ—‰—,—µQR (QReader —¿Ñ—, —‰—°—»—,Ñ—,—,, —,—‰—ºÑ—µOpenCV) + —¿—ºÑÑ—,—‰—‡ Ñ
—⁄— —¡.
- system_prompt.txt: Ñ—,ÑÑ—µ—…—‰Ñ—„ —¿Ñ—¾—…—¿Ñ (ÑÑÑ—¾—‡—,—„ JSON, —¿—¾—»—‰—¾Ñ—° —¿—¾—•—,Ñ—,
Ñ—°—…—¾—¿Ñ—¾—†—µÑ—”—º).
- google_sheets.py: —•—°—¿—,ÑÑ —† Google Sheets + —†—°—»—,—·—ºÑ—,—,/—¿Ñ—µ·Ñ—¿Ñ—µ—¶—·—µ—‰—,Ñ, —”

— —¾Ñ—¾—” —¾—Ñ—°———¾Ñ—”—, —¾·—‰—¾—‡—¾Ñ—µ—”—°
1. — Ñ—µ·—¾—Ñ—°———¾Ñ—”—° —,—•—¾—Ñ—°—¶—µ—‰—,—„ (image_preprocess.py): EXIFâ —¿—¾—†—¾Ñ—¾Ñ, resi
(Ñ—¾—»Ñ—”—¾·—»Ñ —¾·—,—‰—¾Ñ—‰—¾—‡—¾—”—°·Ñ—º).
2. —⁄—,Ñ—”—°—»Ñ—‰Ñ—„ QR (—¿Ñ—,—¾Ñ—,Ñ—µÑ):
   - — —µ—”—¾·—,Ñ—¾—†—°—‰—,—µQR: QReader (—µÑ—»—, ÑÑÑ—°—‰—¾—†—»—µ—‰ —, Ñ—°———¾Ñ—°—µÑ) â —
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—⁄— (—”—°—” â —‰—¾…—µÑ Ñ—µ—”—ºâ), fp, —,Ñ—¾—‡—¾—†—ºÑ ÑÑ—…—…—°.
3. OpenAI Vision (ocr.py):
   - —¡—,ÑÑ—µ—…—‰Ñ—„ —¿Ñ—¾—…—¿Ñ —,—• system_prompt.txt.
   - Userâ —,—‰ÑÑÑÑ—”Ñ—,Ñ —,—• build_user_text() (—·—»Ñ —·—»—,—‰—‰ÑÑ Ñ—µ—”—¾—† â —¿Ñ—¾—¿—¾—»—‰
   - — Ñ—»—, QR Ñ—ºÑ—¿—¾—•—‰—°—‰ â —† userâ Ñ—µ—”ÑÑ —·—¾——°—†—»Ñ—µÑÑÑ —”—¾—‰Ñ—µ—”ÑÑ Ñ ÑÑ—°
4. —¡—»—,Ñ—‰—,—µÑ—µ—•Ñ—»ÑÑ—ºÑ—°:
   - — —¾—»Ñ —,—• QR —¿Ñ—,—‰Ñ—·—,Ñ—µ—»Ñ—‰—¾—¿—ºÑÑ—ºÑÑÑ —† receipt_info
(date/time/receipt_number/fiscal_sign/total_amount/fiscal_storage_number).
5. — —°—»—,—·—ºÑ—,Ñ / —¿—¾—†Ñ—¾Ñ:
   - should_recheck() Ñ—µÑ—°—µÑ, —‰Ñ—¶—‰—° —»—, 2â Ñ —¿—¾—¿ÑÑ—”—° LLM.
   - _meta.needs_review —†ÑÑÑ—°—†—»Ñ—µÑÑÑ —¿—¾Ñ—,—‰—°—»Ñ—‰—¾…Ñ JSON —¿—¾Ñ—»—µ—¿—¾—¿ÑÑ—¾
6. — —°—¿—,ÑÑ —† Sheets:
   - — —,ÑÑ Ñ—µ—”—¾—†: Aâ N ÑÑ—°—‰—·—ºÑÑ—‰Ñ—µ—¿—¾—»Ñ, O = —⁄— (—,—• QR), P = QR Ñ—ºÑ—¿—¾—•—‰—
   - — —,ÑÑ Ñ—¾—†—ºÑ—¾—†: —¿—¾—•—,Ñ—,—, items.